DATA-Snowflake-IICS-Tableau

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Week	Day	Topic & Skill	Content Coverage
1	1	Programming Logic and Techniques (PLT)	**Flow Charts –** • Definition and importance of flow charts, Real-world applications of flow charts. • Common flowchart symbols (Start/End, Input/Output, Process, Decision), Creating simple flow charts. • Step-by-step approach to designing flow charts, Practice session: Creating flow charts for basic tasks. **Algorithms –** • Definition, Importance, and Types of algorithms (Sequential, Conditional, Iterative). • Identifying inputs/outputs, Breaking down problems into steps, Example algorithms (Sorting, Searching). • Writing algorithms for simple tasks (e.g., finding factorial, sorting numbers). **Pseudo Code –** • Definition, Purpose, and Basic structure of pseudo code. • Translating flow charts into pseudo code, Writing simple pseudo code for algorithms. • Writing pseudo code for common problems (e.g., Fibonacci series, prime number check).
	2	Introduction to Web Technology, HTML, Basic Tags and CSS	**Html** • Html • Body • Head • Title • P • Div • Ul, li , ol • H1 to h6 • B, I, u • Br, hr • Form • Table • Img, audio, video • Anchor tag **Introduction of CSS** • How to connect CSS with html: inline, internal, external • Basic CSS properties: width, height, background, border, font, foreground, padding, margin, color
	3	Introduction to Web Service	• What are web services • Web service architecture • REST vs SOAP • Swagger Docs for REST APIs • Http methods • Demo with Swagger pre-defined REST endpoints

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Week	Day	Topic & Skill	Content Coverage
	4	Introduction to Cloud Computing and Python basics	**Introduction to Cloud Computing** • What is cloud computing • Why cloud computing • Examples of Cloud Computing (IaaS, SaaS, PaaS) • Types of cloud computing (private, public, hybrid, community) • Cloud Service Provider (AWS, Azure, GCP) **Introduction to Python** • Input output statements in python (print, input) • Print function with keyword arguments (sep, end) • Data types in python (str, int, float, complex) • Expression • Operators • Type casting • Conditional statements • Looping statements • Jumping statements • Some special functions o Len o Id o Type o range+
	5	Python: List, Tuple, and Set	• Introduction to Data structure in python • List in python o Introduction to List with Properties o List methods _Append, Insert, Remove, Copy (shallow and deep copy), Count, Extend, Index, Sort (ascending and descending), Reverse, Clear, Pop o List comprehensions • Indexing o Positive indexing, Negative indexing, Retrieve, update • Slicing o Positive Slicing, Negative Slicing, Retrieve, Update, Delete, insert • Searching and sorting with list • Introduction to tuple o Properties of tuple o List vs tuple o Indexing and slicing on tuple o Tuple specific methods: count, index • Python in-build functions with List and Tuple o sum o max o min o avg o len o sort • Convert string into tuple and list • Tuple and list specific programs • Set : data structure o Introduction to Sets • Introduction to sets: Unordered collection of unique elements. • Creating sets: set() constructor, curly braces {}. • Basic operations: Union, intersection, difference, symmetric difference. • Practical examples and use cases.

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Week	Day	Topic & Skill	Content Coverage
			o Set Methods and Operations • Set methods: add(), remove(), discard(), clear(), etc. • Checking membership with in keyword. • Set comprehension. • Advanced set operations and examples. • Dictionaries in Python o Introduction to Dictionaries • Introduction to dictionaries: Key-value pairs. • Creating dictionaries: dict() constructor, curly braces {}. • Accessing and modifying values using keys. • Practical examples and use cases.

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Week	Day	Topic & Skill	Content Coverage
	6	Python: Dictionary, User Defined Functions	o Dictionary Methods and Operations • Dictionary methods: keys(), values(), items(), etc. • Adding and removing key-value pairs. • Dictionary comprehension. • Nesting dictionaries and practical examples. o Use Cases and Real-world Examples • Solving problems using sets and dictionaries. • Examples from data manipulation, filtering, and analysis. • Best practices for choosing between sets and dictionaries. • Function in Python o Introduction to function o Syntax and examples o Local and global variables o Global keyword • Parameters in function Positional arguments, Keyword arguments, Positional only arguments, Keyword only arguments, Default arguments, Variable length arguments, Keyword Variable length arguments, Pass function as an argument • Lambda function • Map, reduce and filter • Global and local variables • Return keyword in function Return a value, Return multiple values, Return function **Weekly Assessment-1**
2	1	SQL: Basics of DBMS, DDL, DML, DQL and Operators	• Introduction to MySQL • Datatypes • Numeric • Date and time • String • DDL – DML commands • Create • Drop • Alter • Truncate • Constraints • Unique • Not null • default • Primary key • Foreign key • Check • DML – DML Commands • Update • Delete • insert

Week	Day	Topic & Skill	Content Coverage
	2	SQL: Build-in functions (str,math), group by order by	• Operators • Arithmetic (+, -, /, %, *) • Bitwise (&, |) • Comparison (<, >, =, <=, >=, <>) • Logical AND, ALL, ANY, BETWEEN, EXISTS, IN, LIKE, NOT, OR, SOME • DQL • Select query • Group by clause • Order by clause • In-build functions • String functions Char length, Ascii, concat, Instr, Lcase, Ucase, Substr, Lpad, Rpad, Trim • Numeric functions Avg, Count, Pow, Min, Max, Round, Sqrt, Floor • Date and time functions Current_date, Day_diff, Date, Current_time, Last_day, Sys_date, adddate
	3	SQL: Joins and Normalization	• Joins • Inner join • Left join • Right join • Full outer join • Cross join • Self join • Normalization • Anomalies • Keys (candidate keys) • Primary key and foreign key constraints • 1st Normal Form • 2nd Normal Form • 3rd Normal Form • Boyce cod normal form
	4	**Mock Test-1 + Debriefing**	
	5	**Mock Test-2 + Debriefing**	
	6	**Milestone 1 Assessment + pre-CEFR**	
3	1	SQL: Complex Joins, Subqueries, Date and Time functions	• Subqueries • Single row subqueries • Multi row subqueries • Correlated subqueries queries • Joins • Joins with subqueries • Joins with aggregate functions • Joins with date and time functions
	2	SQL: Analytical and Advanced Functions	• RANK • DENSE_RANK • ROW_NUMBER • CUME_DIST • LAG • LEAD

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Week	Day	Topic & Skill	Content Coverage
	3	SQL: Procedures, Functions and Cursor	• Procedures • Functions • IN • OUT • INOUT • Cursors User defined, Pre-defined **Milestone - 1 Retest (if any)**
	4	SQL: TCL, Triggers and Views	• TCL commands • Commit • Rollback • Save point • Triggers - using insert, update, delete operations • View • Creating/replacing views • Dropping views
	5	NoSQL and MongoDB	• Introduction to NoSQL o Overview of relational databases vs NoSQL databases. o Types of NoSQL databases: Document-oriented, Key-Value, Column-family, Graph databases. o Characteristics of NoSQL databases: Schema-less, scalability, flexibility. o Use cases for choosing NoSQL databases. • Introduction to MongoDB • Overview of MongoDB: Document-oriented database. • MongoDB's JSON-like BSON format. • Collections and Documents: Equivalent to tables and rows in relational databases. • Installation of MongoDB locally. • MongoDB CRUD Operations • Inserting documents. • Querying documents. • Updating documents. • Deleting documents. • MongoDB Data Modeling • Introduction to BSON data types. • Schema design best practices. • Embedded documents and arrays. • Relationships and normalization in MongoDB. • Indexing and Query Optimization • Importance of indexing. • Creating and managing indexes. • Query optimization techniques. • Explain plan and performance monitoring. • Introduction to MongoDB Atlas • Overview of MongoDB Atlas. • Creating a free MongoDB Atlas account. • Deploying a cluster on MongoDB Atlas. • Aggregation Framework in MongoDB • Overview of the aggregation pipeline. • Using $match, $group, $project, and other stages. • Aggregation examples for data transformation.

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Week	Day	Topic & Skill	Content Coverage
	6	Linux Basics and Linux	• Linux Fundamentals -Overview • What is Linux and Linux Distributions • Components of Linux System • Basic Features of Linux System • Linux File System • Bash and PowerShell Scripting • Daemons • Graphical Servers and Desktop Environment • Applications • What is Powershell • BootLoader • Linux Kernel • Linux Commands related to editor • Linux commands related to version control • Linux General Commands • SED editor
4	1	Command Line Programming using Linux	• Ability to obtain data using command line from various resources like CSV, JSON, XML etc.. • Ability to create reusable command line tools • Ability to apply scrubbing operations using command line operators • Ability to manage data using command line operators • Ability to explore data using command line operators • Ability to work with parallel pipelines using command line operators **Weekly Assessment-2 (SQL, Linux Basics and Linux)**
	2	GIT + DevOps	• Learning Objective Git and Agile • Introduction to Version Control Systems • Introduction to Git • GIT Installation • Basic Terminal Commands • GIT init • GIT config • Tracked and Untracked Files • GIT log • GIT commit • GIT checkout • GIT reset • GIT ignore • GIT add and commit • Introduction to Github • Building Github repository • Pushing local repository to Github • Commit and Pull request in Github • Branching in Github • Merge conflict • Git Rebase • Merging branches and resolving conflict in Github • Learning outcomes- Git

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Week	Day	Topic & Skill	Content Coverage
	3	Python: Revision of Function, Class, Object, General OOPs concept and OOPs in Python	• Revision of data structure in python (List, Set, Tuple, Dict) • Revision of Function in python **Objects & Class** • OOPs in python • Introduction to OOPs principles • General OOPs vs Python OOPs • Class and object • Data encapsulation • Abstraction • Data hiding • Methods in class Normal methods, Class methods, Static methods, Data members in class, Instance variables, Class variables • Methods with arguments Pass object as an argument, Object as an return type, Method overloading
	4	Python: Constructor, Destructor, Inheritance	• Special methods __init__ __str__ __new__ • Constructor and destructor in python Constructor with default arguments Constructor without arguments Constructor with arguments Destructor: Define and invocation of destructor • Inheritance in python Introduction and properties of inheritance MRO (method resolution order) Types of inheritance Constructor and destructor with inheritance • Method overriding Method overriding Super method Super method with special methods (__init__,__str__)

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Week	Day	Topic & Skill	Content Coverage
	5	Python: Exception Handling, SQL & NoSQL Connectivity	● Exception Handling • Introduction to Exceptions ▪Definition of Exceptions ▪Why Exception Handling is important • Basics of Exception Handling ▪Syntax of try and except blocks ▪Handling multiple exceptions ▪Using else and finally blocks • Common Built-in Exceptions ▪SyntaxError ▪IndentationError ▪NameError ▪TypeError ▪ValueError ▪ZeroDivisionError ▪Custom exceptions (brief introduction) • Handling Specific Exceptions ▪Using except with specific exception types ▪Using as to get exception details • Raising Exceptions ▪Using the raise statement • Exception Hierarchy ▪Understanding the hierarchy of exceptions in Python ▪Handling exceptions at different levels • Best Practices for Exception Handling ▪Properly documenting code with exceptions ▪Avoiding broad except clauses ▪Using finally for cleanup operations • Practical Examples and Exercises ▪Writing code snippets to demonstrate exception handling ● Python SQL Connectivity • Introduction to Database Connectivity Briefly explain the importance of connecting Python to databases in real-world applications. ▪Introduce common use cases such as data storage, retrieval, and manipulation. • Overview of Database Systems ▪Discuss different types of database systems (e.g., SQLite, MySQL, PostgreSQL). ▪Highlight the strengths and use cases of each database system. ▪Emphasize the importance of choosing a database system based on project requirements. • Explain / Installing Database Drivers ▪Guide participants through the installation of a popular database driver such as sqlite3, mysql-connector, or psycopg2. ▪Explain how to install drivers using package managers like pip. • Establishing Database Connection ▪Demonstrate how to establish a connection to a database using Python. ▪Discuss connection parameters such as hostname, username, password, and database name. ▪Provide examples for both local database connections. • Executing SQL Queries ▪Introduce the concept of SQL queries in Python. ▪Show how to execute SELECT, INSERT, UPDATE, and DELETE queries using Python. • Fetching and Processing Results ▪Demonstrate how to fetch and process the results of SQL queries in Python.

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Week	Day	Topic & Skill	Content Coverage
			Discuss different methods for fetching results (fetchone, fetchall, fetchmany). ▪Cover techniques for iterating through query results and processing data. • Error Handling ▪Common errors that might occur during database operations. ▪How to handle errors gracefully using try-except blocks. ▪Emphasize the importance of error handling for robust database applications. • Hands-On Exercise ● Python NoSQL / MongoDB Connectivity • Introduction to MongoDB ▪Recap of MongoDB • Installing PyMong• ▪Install PyMong• using pip. ▪Verify the installation and version. • Connecting to MongoDB ▪pymongo.MongoClient class for establishing a connection. ▪connection parameters, including the MongoDB server URI. ▪Demonstration of how to connect to a MongoDB server. • Creating a Database and Collection ▪Concept of databases and collections in MongoDB. ▪Demonstration of how to create a new database and collection using PyMongo. ▪Demonstration of handling databases and collections that don't exist yet. • Inserting Documents ▪insert_one and insert_many methods for inserting documents. ▪Discussion of the structure of MongoDB documents (JSON-like format). ▪Demonstration of inserting single and multiple documents. • Querying Documents ▪find method for querying documents. ▪various query operators and projection options. ▪filtering, sorting, and limiting query results. • Updating Documents ▪update_one and update_many methods for updating documents. ▪$set operator and other update operators. ▪updating documents based on specific criteria. • Deleting Documents ▪delete_one and delete_many methods for deleting documents. ▪cascading deletes. ▪deleting documents based on specific criteria. • Error Handling ▪common errors that might occur during MongoDB operations. ▪handle errors gracefully using try-except blocks. • Hands-On Exercises

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Week	Day	Topic & Skill	Content Coverage
	6	Python: PyTest, Introduction to NumPy	● Pytest ● Introduction to Testing • Briefly explain the importance of testing in software development. • Different types of testing (unit testing, integration testing, etc.). • Benefits of automated testing. ● Setting Up Pytest • Install Pytest using pip. • Directory structure for test files and the naming convention. • Organize test files and test functions. ● Writing Your First Test • Basic structure of a Pytest test function. • Demonstrate writing simple test functions for basic assertions. • Common assertion methods provided by Pytest. ● Hands-On Exercise ● NumPy • Introduction to NumPy ▪NumPy as a fundamental library for numerical operations in Python. ▪Benefits of using NumPy over standard Python lists. • Installing NumPy ▪Demonstrate how to install NumPy using pip. ▪Verify the installation and version. ● NumPy Arrays • Introduce NumPy arrays as the core data structure. • Discuss the advantages of NumPy arrays over lists. • Demonstrate creating arrays using np.array() and exploring array attributes (shape, size, dtype). • Other NumPy functions ▪asarray(), ones(), zeros(), empty()

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Week	Day	Topic & Skill	Content Coverage
5	1	Python: NumPy, Pandas	NumPy ● Array Operations and Broadcasting • Explore basic array operations (addition, subtraction, multiplication, etc.). • Broadcasting for performing operations on arrays of different shapes. • Element-wise operations and their efficiency. • Indexing and Slicing ▪Indexing and slicing in NumPy arrays. ▪Boolean indexing ▪Fancy indexing ▪Copy of ndarray • Array Manipulation ▪Discuss reshaping, flattening, and concatenating arrays. ▪Introduce stacking and splitting of arrays. ▪Explore methods for adding and removing elements. • Hands-On Exercises Pandas • Introduction to Pandas ▪Importance of data manipulation in data science and analysis. ▪Pandas a library for handling structured data in Python. ▪Benefits of using Pandas for data cleaning and analysis. • Installing Pandas ▪Demonstrate how to install Pandas using pip. ▪Verify the installation and version. • Pandas Series ▪Series an one-dimensional labeled arrays. ▪Creating Series from lists, arrays, and dictionaries. ▪Explore Series attributes and methods for data manipulation. • Pandas DataFrame • Pandas DataFrames a two-dimensional labeled data structures. • Creating DataFrames from dictionaries, lists, and external files. • Explore DataFrame attributes, indexing, and basic operations.
	2	Python: Pandas	• Data Inspection and Exploration • Methods for inspecting data, such as head(), tail(), and info(). • Importance of understanding the structure and types of data. • Descriptive statistics using describe(). • Data Cleaning with Pandas • Common data cleaning tasks, such as handling missing values. • Methods for handling missing data, including dropna() and fillna(). • Handle duplicates and outliers. • Indexing and Selection in DataFrames • loc and iloc for label-based and integer-based indexing. • slicing • boolean indexing • Boolean indexing for filtering data. • Data Manipulation and Transformation • Methods for reshaping and transforming data, including groupby() and pivot_table(). • Demonstrate how to apply functions to data using apply() and map(). • Explore merging and concatenating DataFrames. • Hands-On Exercises **+ Weekly Assessment-3**

Week	Day	Topic & Skill	Content Coverage

	3	Python: Visualisation - Matplotlib & Seaborn	• Introduction to Data Visualization
			  • Importance of data visualization in conveying information.
			  • Introduce key concepts such as data types (categorical, numerical) and chart types (scatter plots, bar charts, etc.).
			  • Role of visualization in data analysis and storytelling.
			• Installing Matplotlib and Seaborn
			  • Install Matplotlib and Seaborn using pip.
			  • Importance of using virtual environments for dependency management.
			  • Verify the installation and version.
			• Matplotlib
			  • Introduction to Matplotlib
			  ▪Matplotlib as a versatile plotting library.
			  ▪Create basic plots (line plots, scatter plots) using pyplot.
			  ▪The anatomy of a Matplotlib figure (axes, labels, title).
			  • Customizing Matplotlib Plots
			  ▪Explore customization options for plots, including colors, markers, and line styles.
			  ▪Set titles, labels, and legends.
			  • Working with Different Types of Plots
			  ▪Line Plot
			  ▪Scatter Plot
			  ▪Bar Plot
			  ▪Histogram
			  ▪Box Plot
			  ▪Pie Chart
			  ▪Area Plot
			• Seaborn
			  • Introduction to Seaborn
			  ▪Role of Seaborn in statistical data visualization.
			  ▪Seaborn's integration with Matplotlib and Pandas.
			  ▪Aesthetics in Seaborn.
			  • Installing Seaborn
			  ▪Install Seaborn using pip.
			  ▪Importance of using virtual environments for dependency management.
			  ▪Verify the installation and version.
			  • Basic Seaborn Plots
			  ▪Explore basic Seaborn plots: sns.scatterplot, sns.lineplot, and sns.barplot.
			  ▪Discuss the simplicity and enhanced aesthetics provided by Seaborn.
			  ▪Comparison between basic Seaborn plots with their Matplotlib equivalents.
			  • Distribution Plots
			  ▪Introduce distribution plots: sns.histplot, sns.kdeplot, and sns.rugplot.
			  ▪Use of distribution plots for understanding data distributions.
			  ▪Demonstrate customizations for better readability.
			  • Categorical Plots
			  ▪Explore categorical plots: sns.boxplot, sns.violinplot, and sns.swarmplot.
			  ▪Visualization of categorical data and the distribution within categories.
			  • Pair Plots and Heatmaps
			  ▪Introduce pair plots (sns.pairplot) for visualizing relationships between multiple variables.
			  ▪Discuss the use of heatmaps (sns.heatmap) for displaying correlations in a matrix.
			  • Hands-On Exercises

	4	**Mocktest-1 and Debriefing**	
	5	**Mocktest-2 and Debriefing**	
	6	**Milestone 2 Assessment**	

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Week	Day	Topic & Skill	Content Coverage
6	1	Snowflake: Introduction and Data Loading (Introduction to Snowflake and Basic Configuration)	● Key Concepts & Architecture ● Supported Cloud Platforms ● Supported Cloud Regions ● Snowflake Editions ● Snowflake Releases ● Overview of Key Features ● Overview of the Data Lifecycle ● Continuous Data Protection ● Snowflake Ecosystem ● Snowflake Partner Connect ● General Configuration (All Clients) ● SnowSQL (CLI Client) ● Connectors & Drivers ● Overview of Data Loading ● Summary of Data Loading Features ● Data Loading Considerations ● Preparing to Load Data ● Bulk Loading Using COPY ● Loading Continuously Using Snowpipe ● Loading Using the Web Interface (Limited) ● Querying Data in Staged Files ● Querying Metadata for Staged Files ● Transforming Data During a Load
	2	Snowflake: Data Unloading and Querying (Advanced Data Loading, Querying and Web Interface)	● Overview of Data Unloading ● Summary of Data Unloading Features ● Data Unloading Considerations ● Preparing to Unload Data ● Unloading into a Snowflake Stage ● Unloading into Amazon S3 ● Unloading into Google Cloud Storage ● Unloading into Microsoft Azure ● Classic Web Interface ● New Web Interface ● Virtual Warehouses ● Databases, Tables & Views ● Queries ● Binary Data ● Date & Time Data ● Semi-structured Data ● Snowflake Time Travel & Fail-safe ● Continuous Data Pipelines (SnowPipe) ● Database Replication and Failover/Failback

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Week	Day	Topic & Skill	Content Coverage
	3	Snowflake: Data Sharing and Management for Organization (Secure data sharing and organizing)	● Introduction to Secure Data Sharing ● Overview of the Product Offerings for Secure Data Sharing ● Granting Privileges to Other Roles ● Working with Shared Data ● Secure Direct Data Share ● Snowflake Data Marketplace ● Data Exchange ● Introduction to Organizations ● Getting Started with Organizations ● Managing Accounts in Your Organization ● Understanding Organization and Account Names **Milestone-2 Retest (if any)**
	4	Snowflake: Data Security, Governance and Application Development (Snowflake Security and SnowSQL)	● Summary of Security Features ● Authentication ● Networking & Private Connectivity ● Administration & Authorization ● Summary of Governance Features ● Column-level Security ● Row-level Security ● Access History ● Download and Install the SnowSQL CLI Client ● Create a Virtual Warehouse, Sample Database, and Table Using SnowSQL ● Load and Query Data Using SnowSQL ● Use Python with Snowflake
	5	Snowflake: Managing Accounts-1 (Snowflake Resource Management)	● System Usage & Billing ● Understanding Snowflake Credit and Storage Usage ● Understanding Snowflake Data Transfer Billing ● Monitoring Account-level Credit and Storage Usage ● Working with Resource Monitors ● Parameter Management ● User Management ● Behavior Change Release Management ● Monitoring and Status Page for Snowflake ● Managing Multiple Snowflake Accounts ● Enabling Data Sharing Within Your Modern Data Architecture ● Integrating and Streaming Data ● Integrating Data Into Snowflake Batch and Streaming ● Ingesting Streaming Data Continuously Using Snowpipe

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Week	Day	Topic & Skill	Content Coverage
	6	Snowflake: Managing Accounts-2 (Advanced topics and best practices)	● Get the Most Out of Snowflake's Partner Ecosystem ● Data Warehouse Modernization ● Getting Started with Snowflake - The Architecture ● Micro Partitions: The Secret to Zero-Copy Cloning, Time Travel, and Continuous Data Protection ● Snowflake Optimizer and Query Profile ● Elasticity, Concurrency & Cost Control in Snowflake ● Migrating to Snowflake with Zero Disruption ● Snowflake Your Data Fast, Easy BI with Snowflake ● Get the Most Out of Snowflake for BI and Analytics ● Data and Decision Science at Scale Technology and Culture Shift ● How to Move from HIVE Hadoop Spark Infrastructure to Snowflake ● Discover Insights from Your Snowflake Usage Data ● Rapid Scaling with Snowflake ● Beyond the Data Warehouse ● Snowflake as a Data Lake ● Overview of AI, ML, and Data Science in the Snowflake Ecosystem ● Many Pools to Just One Modern Data Lake ● Deep Dive on Multi-Cloud External Stages in Snowflake ● Snowflake Coding Standards ● Snowflake Best Practices ● Snowflake Reference Architectures ● Use Cases - Snowflake Implementations
7	1	IICS: Cloud Data Integration Service-1 (IICS Modules)	● Describe Informatica Cloud Architecture ● Install the secure agent and create connections ● Create Synchronization task ● Create Data Transfer task ● Use Cloud Mapping Designer to create Mappings and Mapping tasks ● Create a Replication task ● Create a Masking task **Weekly Assessment-4 (Snowflake)**
	2	IICS: Cloud Data Integration Service-2 (IICS Modules)	● Create a Mass Ingestion task ● Create Task Flows ● Use IICS REST web services for data integration ● Use Intelligent Structure Model to parse data
	3	IICS: Cloud Data Integration Service-3 (IICS Modules)	● Handle exceptions ● Use advanced data integration features to optimize performance of jobs ● Automate and monitor tasks ● Configure advanced administration settings in IICS
	4	IICS: Cloud Data Integration Service-4 (Users and Asset management)	● Distinguish users and groups ● Configure custom roles in IICS ● Asset Management
	5	IICS: Cloud Application Integration for Developers-1**+Certificate Preparation** (IICS and CAI)	● Explain Informatica Intelligent Cloud Services (IICS) and Cloud Application Integration (CAI) ● Describe Processes and Guides

Week	Day	Topic & Skill	Content Coverage
	6	IICS: Cloud Application Integration for Developers-1**+Certificate Preparation** (Integration and API Management)	● Consume third-party applications and REST/SOAP services ● Establish error handling routines ● Expose developed assets as API endpoints ● Manage APIs
8	1	**IICS: Certification Preparation**	**Weekly Assessment-5 (IICS: Certification)**
	2	Tableau: Overview, basics, first dashboard, Tableau Public (Tableau Modules)	● Overview of Tableau Public and its features ● Understanding the Tableau Public platform ● Connecting to data sources in Tableau Public ● Explore the Tableau Public visualization pane interface ● Building a basic visualization in Tableau Public ● Data Types in Tableau ● Create a Map Chart ● Create a Filled Map ● My First Dashboard ● Publishing visualizations to Tableau Public
	3	Tableau: Filters-Dimension, Measures, Cascading, Hierarchi. Filtering options, Sorting (Filters in Tableau)	● Dimension Filters in Tableau ● Single Value and Multiple Value options in Filters ● Measure Filters in Tableau ● Range and Slider options in Filters ● Cascading Filters in Tableau ● Create a Hierarchy ● Hierarchical Filters in Tableau ● Explaining Relevant, All Members in Column and All members from Filtering options ● Sorting, Nested Sorting and Manual
	4	Tableau: Data fields, Calculated fields, Date Filters, Creating shapes and visualization formatting (Charts in Tableau)	● Handling Date fields in Tableau ● Creating Line Chart ● Applying Date Filters in Tableau ● Formatting visualizations in Tableau Public ● Introduction to conditional calculated fields in Tableau ● Creating a Shape / KPI Chart
	5	Tableau: Conditional Fields, Functions (Functions)	● Conditional Fields using IF and NESTED IF function ● Conditional Fields using CASE function ● String Functions in Tableau ● Number Functions in Tableau ● Statistical Functions in Tableau ● Aggregate Calculations in Tableau
	6	Tableau: Joins and Relationships (Data Modeling)	● Introduction to Joins and Relationships ● Introduction to Groups and Sets ● Creation of Static, Dynamic and Combined Sets ● Introduction to Bins

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Week	Day	Topic & Skill	Content Coverage
9	1	Tableau: Plots and Charts (Visualization in Tableau)	● Building Scatter Plots in Tableau ● Creating dual-axis and comb• charts in Tableau Public ● Implementing trend lines and forecasting in Tableau Public ● Creating Pie Charts, Word Maps, etc ● Understanding table calculations in Tableau Public ● Creating Waterfall Charts, Bubble Charts, Custom Maps, etc. ● Detailed case study with Tableau
	2	**Mocktest-1 and Debriefing**	
	3	**Mocktest-2 and Debriefing**	
	4	**Milestone 3 Assessment**	
	5	**Mock client Interview**	
	6	Capstone Project	
10	1	Capstone Project + **Milestone-3 Retest (if any)**	
	2	Capstone Project	
	3	Capstone Project	
	4	Capstone Project	
	5	Capstone Project	
	6	Capstone Project preparation	
11	1	**Milestone - 4: Capstone Project Evaluation**	

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